Fiscal Year:	FY 2021	**Task Last Updated:**	FY 03/18/2024
PI Name:	Hayman née Anderson, Allison Ph.D.		
Project Title:	Performance Enhancement Through Multi-Modal Stochastic Resonance		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: name change to Hayman née Anderson (Ed., March 2025). PI moved to University of Colorado from Dartmouth College in early 2017.		
Project Type:	Ground	**Solicitation / Funding Source:**	2018 TRA BRASH1801: Translational Research Institute for Space Health (TRISH) Biomedical Research Advances for Space Health
Start Date:	01/01/2019	**End Date:**	12/31/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	10	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2021 per E. Urquieta/TRISH (Ed., 4/14/21) NOTE: End date changed to 5/31/2021 per TRISH Task Book submission (Ed., 7/2/2020)		
Key Personnel Changes/Previous PI:	Ed. note 7/2/2020--CoInvestigators Mulavara and Bloomberg are no longer working on the project.		
COI Name (Institution):	Clark, Torin Ph.D. (University of Colorado, Boulder)		
Grant/Contract No.:	NNX16AO69A-T0402		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This project's objective is to investigate the operational benefit of brain stimulation with stochastic resonance (SR). The team will measure performance enhancement across multiple sensory modalities, in complex tasks, and over repeated use when stochastic resonance is applied through the auditory and vestibular sensory pathways. To enhance neurocognitive and task performance, stochastic resonance may be used to enhance information transfer across multiple sensory modalities, reducing the cognitive workload associated with complex tasks. SR is achieved when sub-threshold white noise is added to a perceptual channel (e.g., auditory, vestibular, visual, etc.), and resonates with the underlying primary signal. As a result, the signal's message is enhanced, allowing the human to perceive the primary signal more easily. Not only does SR increase performance on the modality to which it is applied, it also shows cross-modality benefits and improves perception in other sensory systems. For example, adding auditory white noise can improve the ability to sense touch or light. Further, by improving perception, there may be secondary benefits of SR such as improved memory, enhanced sleep, and reduced stress.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research investigates the use of SR as a noninvasive, easy to administer way to stimulate the brain and improve performance in spaceflight relevant tasks. It investigates multimodal SR through the auditory and vestibular channels without requiring bulky, difficult to wear, high power devices that cannot be used in operational environments. This research incorporates an evaluation of the long-term effects of SR in an operational spaceflight analog. SR is likely to be highly acceptable to astronauts because it is applied in a manner that is non-invasive, stimulates multi-sensory neurons in the brain through natural sensory pathways, and does not have the harmful or unpredictable outcomes seen in transcranial magnetic or current stimulation. Multi-modal SR has not previously been investigated nor applied to complex spaceflight relevant tasks. Because SR stimulates the brain and improves performance across all sensing modalities, we expect it to show improvements across a broad range of tasks. These improvements in performance achieved through SR may be critical for times of high stress, where resiliency is required and improvement in performance can make the difference, for example during vehicle launch, landing, or emergency situations. It may also show long-term benefits by reducing the cognitive demands on every-day tasks, stimulating the brain and preserving brain function. In Year 1, our research has identified novel findings (vestibular stochastic resonance for improvement in the visual perception task), confirmed findings in the literature (auditory stochastic resonance for the auditory task), and has developed novel experimental design methodologies. It has furthered our ability to objectively assess SR, adding to the literature to ensure these improvements are rigorously identified. We have begun the process of validating this work in a spaceflight relevant analog by successfully being integrated with the NASA Human Exploration Research Analog (HERA) Campaign 6. One additional area of impact not originally identified but that was added opportunistically to this study was to investigate the use of SR while performing the Cognition Test Battery, which is being done by all researchers within our cohort. We have begun the initial experimental design and will perform this assessment in Year 2. This research addresses the Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorder. This research is related to gap BMed1: we need to identify and validate countermeasures that promote individual behavioral health and performance during exploration class missions. The proposed research investigates the effects of SR on performance in spaceflight relevant tasks by increasing attention, perception, and information integration, with the potential to provide secondary benefits in memory, stress, and mood.
Task Progress:	This research explored stochastic resonance (SR) as an approach to enhance perception, neurocognition, operational task performance, learning, and secondary benefits such as mood or stress. SR is a phenomenon in which white noise is added to a perceptual channel (e.g., auditory or vestibular), which resonates (in the sensory and central neural circuitry) to improve “information transfer” to and within the brain. As an example of “in-channel” SR, noisy galvanic vestibular stimulation (nGVS) has been shown to improve vestibular self-motion perceptual thresholds. Expanding upon this, we explore “cross-modal” SR in which white noise is applied in one sensory channel (e.g., nGVS) to improve perceptual thresholds in another channel (e.g., visual thresholds). Finally, “multi-modal” SR refers to white noise being administered in multiple sensory channels (nGVS and auditory white noise, or AWN). Through our four experimental investigations and the development of novel experimental design methods to increase statistical rigor, we perform one of the most comprehensive analysis of the effects of SR on human performance. In our human subject experiments, we administer nGVS through electrodes placed on the mastoid processes and AWN through headphones or earphones. When white noise is applied, SR may occur when an optimal low level of white noise is applied; too much white noise will no longer be beneficial and may degrade performance, with the optimal level varying between individuals. Thus, we assess performance enhancement using a range of SR noise levels, fit a characteristic SR-response curve to estimate each subject’s optimal level, and perform further testing near this optimal SR noise level, for both AWN and nGVS. In addition to AWN and nGVS conditions, the blinded subject is tested with a Sham condition and multi-modal SR (MMSR) conditions in which both AWN and nGVS are administered simultaneously. Our Aim 1 investigated perceptual thresholds for auditory, visual, and tactile channels using two-alternative forced choice tasks. To rigorously identify the presence of SR and prevent false positives, we trained a machine learning algorithm using simulated datasets in which the ground truth of the presence of SR is known. We found significant evidence of SR performance improvement in some, but not all individuals for all perception thresholds investigated. Our Aim 2 investigated performance for an operationally relevant task, ground-based planetary landing cockpit simulator. No statistically significant differences were found by SR condition. Our Aim 3 investigated the cognitive benefits of AWN, nGVS, and MMSR. Optimal SR levels were identified and then assessed by performing the Cognitive Test Battery. A statistically significant interaction was found between individuals and SR conditions, and there may be a correlation between subjects self-identified as preferring to work in noisy environments and cognitive performance enhancement with SR. Our Aim 4 investigates the acceptability of SR and potential secondary benefit of SR over repeated use. We also investigate the effect of SR on learning. Data analysis is ongoing, but initial trends suggest that there are no statistical changes of mood and stress with SR and the countermeasure is acceptable to subjects. This TRISH-funded study investigated SR performance enhancement in the areas of perception, cognition, operationally relevant tasks, learning, and secondary benefits. SR may serve as a noninvasive, easy-to-administer method to stimulate the brain and improve performance in spaceflight relevant tasks through the auditory and vestibular channels. We synthesized all our experimental results to assess the degree to which SR is an appropriate countermeasure for human spaceflight performance degradation. We also provide commentary on how SR may best be used, for example in times

	of high stress, where resiliency is required and improvement in performance can be critical (e.g., vehicle launch, landing, or any emergency situations). This research advances the state of SR by providing objective, rigorous methods to investigate improvements in perception and cognition for human spaceflight.
Bibliography Type:	Description: (Last Updated: 02/18/2026)
Articles in Peer-reviewed Journals	Sherman SO, Shen Y-Y, Gutierrez-Mendoza D, Schlittenhart M, Watson C, Clark TK, Anderson AP. "Additive sensory noise effects on operator performance in a lunar landing simulation." Aerosp Med Hum Perform. 2023 Oct;94(10):770-9. https://doi.org/10.3357/AMHP.6251.2023 ; PMID: 37726913 , Oct-2023
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Awards	Voros J. "Zonta International Amelia Earhart Fellowship, University of Colorado." Aug-2020
Awards	Callas M. "Brooke Owens Fellowship, University of Colorado." May-2021
Awards	Lewis Q. "Summer Program for Undergraduate Research, University of Colorado." May-2021
Awards	Rise R. "NSF Graduate Research Fellowship, University of Colorado." Feb-2021
Awards	Anderson, A. "Junior Faculty of the Year, College of Engineering and Applied Sciences, University of Colorado." Apr-2021
Awards	Anderson A. "Junior Faculty of the Year, Smead Department of Aerospace Engineering Sciences, University of Colorado." Feb-2021
Awards	Anderson A. "Outstanding Undergraduate Teaching and Mentorship Award, University of Colorado." May-2021
Awards	Anderson A. "Research and Innovation Office Faculty Fellow, University of Colorado." Jan-2020
Awards	Anderson A. "Young Professional Engineer of the Year, AIAA Rocky Mountain Section, University of Colorado." Jul-2021
Awards	Watson C, Schlittenhart M. "Discovery Learning Apprenticeships, University of Colorado." Aug-2020
Awards	Watson C. "National Undergraduate Research Award, University of Colorado." Jan-2021
Dissertations and Theses	Rise R. "Identifying stochastic resonance in perceptual thresholds with auditory and vestibular white noise." Master's Thesis, University of Colorado, May 2020. , May-2020
Significant Media Coverage	Voros J. (TEDx Talk). "The role of diversity in the next human spaceflight revolution, November 2020." https://www.ted.com/talks/jamie_voros_the_role_of_diversity_in_the_next_human_spaceflight_revolution , Nov-2020